
Experiment-Calibrated Media Mix Models: Bayesian Fusion of Interventional and Observational Evidence for Marketing Decision-Making

Abstract

Marketing-mix models (MMMs) allocate large advertising budgets across channels yet are fit on observational time series in which spend is correlated with unobserved drivers of revenue and with other channels through shared launch windows. We frame budget allocation as off-policy evaluation under partial identifiability and show how interventional evidence from geo-lift and conversion-lift experiments can be fused with the observational MMM as an additional saturation-curve likelihood term, rather than as a prior on uninterpretable regression coefficients. In a controlled simulation the naive observational model roughly doubles the implied effect of the confounded channel; calibration via lift-test likelihood recovers the channel-level return-on-ad-spend in tight credible intervals around ground truth, and we evaluate it in policy space against an oracle allocator with a sensitivity analysis over the experimental prior. The recipe is a deployable instance of causal data fusion for decision making under unobserved confounding, open-sourced in PyMC-Marketing.

1 INTRODUCTION

A retailer choosing how to split an annual advertising budget across paid search, paid social, video, display, and other channels is solving a sequential decision problem under partial identifiability. The value of any candidate allocation depends on causal effects (the incremental revenue produced by an additional dollar of spend on each channel), which observational time series alone cannot identify whenever spend is correlated with unobserved drivers of the outcome. This is the default rather than the exception: marketers raise budgets in periods of strong demand, react to competitors, and follow seasonal or macroeconomic signals that the model

never sees. The identification problem is not only about unobserved confounders. Media-spend time series are typically highly collinear, because campaigns across channels are launched on the same calendar windows; in strongly seasonal businesses such as food delivery, retail flash sales, or travel, paid search, paid social, video, and display are often activated together around the same demand peaks. This co-activation alone is enough to leave channel-level effects weakly identified even before any latent driver is introduced. We frame the problem as off-policy evaluation in a contextual setting where the logging policy is the marketer’s historical behavior, confounded by these unobserved factors [Dudík et al., 2014].

Standard media-mix models (MMMs) are Bayesian regressions that combine adstock and saturation transformations of channel spend with controls for trend and seasonality [Jin et al., 2017, Salvatier et al., 2016, PyMC Labs, 2024a]. They produce posterior intervals on regression coefficients and on derived quantities such as channel return-on-ad-spend (ROAS). When relevant confounders are unobserved, however, the resulting estimates are biased in a direction that the model itself cannot diagnose, and the bias propagates directly into the chosen allocation. In the controlled simulation of Section 5, a naive observational MMM roughly doubles the channel-level effect of a confounded channel, which is enough to reverse the ranking of channels by ROAS. Goodness of fit on held-out windows is uninformative about this failure [Vincent, 2026a, Orduz, 2026].

Geo-lift experiments, conversion-lift studies, and switch-back designs provide local interventional evidence on a subset of channels at a subset of time points. We treat the resulting lift estimates as observations of the saturation curve at known spend pairs and add them to the MMM as a likelihood term, fusing observational and interventional evidence in a single posterior [Bareinboim and Pearl, 2016, Colnet et al., 2024]. Our contributions are: (i) a structural causal formulation of the standard MMM that makes the identification gap explicit; (ii) a calibration likelihood on incremental ROAS that places experimental evidence on the policy-

relevant causal estimand rather than on a model-internal nuisance coefficient; (iii) an evaluation in policy space, instantiated through a constrained-optimization pipeline that admits both expected-revenue and risk-aware Bayesian objectives, and reporting the revenue gap that calibration closes between a naive policy and the oracle policy that knows the confounder; and (iv) a sensitivity analysis quantifying how decision loss degrades when the experimental evidence is biased or stale. The implementation is open-source in PyMC-Marketing.

2 PROBLEM SETUP AND CAUSAL FRAMING

We index time by $t = 1, \dots, T$. Let $X_t \in \mathbb{R}_{\geq 0}^K$ be the spend vector across K channels, W_t a vector of observed controls (seasonality, calendar effects, promotions), Z_t a vector of unobserved confounders, and Y_t the outcome (revenue or conversions). The structural causal model has edges $Z_t \rightarrow X_t$, $Z_t \rightarrow Y_t$, $W_t \rightarrow X_t$, $W_t \rightarrow Y_t$, and $X_t \rightarrow Y_t$. Following the standard MMM specification [Jin et al., 2017], the outcome equation is

$$Y_t = f_{\text{trend}}(t) + g(W_t) + h(Z_t) + \varepsilon_t + \sum_{k=1}^K \beta_k S_{\lambda_k}(A_{\alpha_k}(X_{k,1:t})), \quad (1)$$

where A_{α_k} is a geometric adstock with decay α_k that captures carryover, S_{λ_k} is a logistic saturation with shape λ_k that captures diminishing returns, and ε_t is Gaussian noise. Adstock and saturation are part of the structural equation, not preprocessing.

The causal contrast of interest is the channel-level intervention effect $\mathbb{E}[Y_t \mid \text{do}(X_{k,t} = x)] - \mathbb{E}[Y_t \mid \text{do}(X_{k,t} = 0)]$. Conditioning on W_t alone does not identify it whenever any component of Z_t remains unobserved: the backdoor path $X_{k,t} \leftarrow Z_t \rightarrow Y_t$ stays open, and the regression coefficient β_k in (1) absorbs both the causal effect and the confounding contribution of Z_t . This is the central practical difficulty of MMM: relevant confounders, such as latent demand, supply shocks, competitor behaviour, and price changes, are pervasive and rarely observed. A second, mechanical source of the problem is multicollinearity in X_t itself: because campaigns are typically launched on the same calendar windows, the spend columns share common variation, enlarging the posterior ridge in coefficient space and amplifying any bias that the unobserved Z_t already introduces.

The decision-maker cares not about β_k but about the channel-level ROAS evaluated at a candidate allocation,

$$\text{ROAS}_k(x) = \frac{\mathbb{E}[Y(x_k) - Y(0)]}{\sum_t x_{k,t}}, \quad (2)$$

which is the same quantity that a geo-lift or conversion-lift experiment estimates. Given a total budget B , an allocator

chooses $\hat{\pi} \in \mathbb{R}_{\geq 0}^K$ with $\sum_k \hat{\pi}_k = B$ to maximize expected revenue under the posterior over $(\alpha_k, \lambda_k, \text{ROAS}_k)$. We measure decision loss as the revenue gap to the oracle allocator,

$$\Delta R(\hat{\pi}) = V(\pi^*) - V(\hat{\pi}), \quad (3)$$

evaluated under the true data-generating process. This is the metric we report: not RMSE on β_k , not coverage of credible intervals, but expected money left on the table.

3 ROAS-PARAMETERIZED FRAMEWORK

The natural Bayesian response to confounding is to use informative priors on β_k , but β_k is a coefficient on a transformed and rescaled spend channel whose scale depends on the adstock decay α_k , the saturation shape λ_k , and any standardization of the data; experiments produce evidence on incremental revenue per unit spend, with no direct mapping to β_k without first knowing the nuisance parameters, so prior elicitation in coefficient space is not even well-defined in practice. Following Wurm et al. [2024], we reparameterize the model so that channel ROAS is a primitive parameter and β_k is derived deterministically. For each posterior draw of $(\alpha_k, \lambda_k, \text{ROAS}_k)$ and given the observed spend trajectory $X_{k,1:T}$, we compute

$$\beta_k = \text{ROAS}_k \cdot \frac{\sum_{t=1}^T X_{k,t}}{\sum_{t=1}^T S_{\lambda_k}(A_{\alpha_k}(X_{k,1:t}))}, \quad (4)$$

so that the posterior over ROAS_k is the object of inference and β_k inherits its scale from the saturation transform. This is a domain-specific instance of decoupling the causal target from the nuisance parameters that link it to the observed data [Evans and Didelez, 2024, Orduz, 2024].

Prior elicitation now operates on a quantity that marketers, finance, and product owners can reason about in orders of magnitude. When several lift experiments are available on the same channel, Wurm et al. [2024] recommend constructing the prior on ROAS_k as a weighted sum of the per-experiment ROAS priors with weights proportional to the experimental media spend; this collapses every experiment into a single channel-level prior and discards the information that each experiment carried about the saturation curve at the spend level where it was actually run. The saturation-likelihood calibration of Section 4 retains that spend-level information by entering each experiment as its own likelihood term anchored at the spend pair where it was measured. The reparameterization alone is not yet calibration: the posterior on ROAS_k still inherits the bias of the observational likelihood whenever Z_t is unobserved, which is what the interventional evidence corrects.

4 CALIBRATION VIA LIFT-TEST LIKELIHOOD

A geo-lift or conversion-lift experiment on channel k produces a tuple $(x, \Delta x, \Delta y, \sigma)$: the baseline spend x on the same scale as the MMM’s input, the spend change Δx delivered by the experiment, the estimated outcome lift Δy , and an uncertainty σ on Δy . The lift is a finite-difference observation of the saturation curve at two known spend levels. We add it to the model as a positive-support likelihood on the magnitude of the finite-difference lift,

$$|\Delta y| \sim \text{Gamma}(\mu_k, \sigma),$$

$$\mu_k = |S_{\lambda_k}(A_{\alpha_k}(x + \Delta x)) - S_{\lambda_k}(A_{\alpha_k}(x))|, \quad (5)$$

using the same adstock and saturation transforms that already appear in (1) and PyMC’s (μ, σ) parametrization of the Gamma distribution. The absolute value lets the likelihood accommodate either sign of measured lift on a single positive-support distribution and matches the default implementation in PyMC-Marketing [PyMC Labs, 2024e]; a Normal alternative is exposed by the same API for use cases where signed lifts and symmetric uncertainty are preferred. Multiple lift tests on the same or different channels enter as independent likelihood terms whose product multiplies the observational likelihood. The full posterior fuses the two sources of evidence and produces a single coherent uncertainty over $(\alpha_k, \lambda_k, \text{ROAS}_k)$ for every channel.

The likelihood form (5) places experimental evidence on the saturation curve, which the regression component cannot identify without it. A single lift test still leaves a one-dimensional ridge in the (λ_k, β_k) plane: the same Δy is consistent with a low-saturation, high-coefficient curve and a high-saturation, low-coefficient curve, an identifiability issue documented by Vincent [2026b]. Two or more lift tests at different spend levels resolve the ridge. In practice, even a single test moves the posterior decisively when combined with the observational data: it pins down the curve at one point and the regression component constrains the rest.

The strength of the calibration is governed by σ . A tight σ places strong weight on the experiment and shrinks the posterior toward the experimental measurement; a loose σ recovers the uncalibrated observational fit. This is the Bayesian analogue of the power-prior dial [Ibrahim and Chen, 2000]: σ interpolates between observational-only and experiment-only inference and admits a transparent practitioner interpretation. Lift tests can also be aggregated hierarchically across geographies or campaigns, with channel-level ROAS_k as a partial-pooling target across studies, in the spirit of multiple randomization designs [Masoero et al., 2025]. The implementation in PyMC-Marketing exposes the calibration through a single `add_lift_test_measurements` call [PyMC Labs, 2024a], taking a dataframe with columns `(channel, x, delta_x, delta_y, sigma)`.

5 EXPERIMENTS

5.1 SYNTHETIC ENVIRONMENT

We simulate $T = 131$ weeks of data with $K = 2$ channels. Adstock is geometric with maximum lag 4 and decays $(\alpha_1, \alpha_2) = (0.3, 0.5)$; saturation is logistic with shapes $(\lambda_1, \lambda_2) = (1.0, 2.5)$; coefficients are $(\beta_1, \beta_2) = (2.0, 1.5)$. The unobserved confounder is $Z_t \sim \text{Gamma}(1, 2/3)$ with effect $\beta_Z = 0.3$ and edges $Z_t \rightarrow X_{1,t}$ (channel 1 is confounded; channel 2 is not) and $Z_t \rightarrow Y_t$. Trend is a cubic-root function of time, and seasonality is a Fourier basis of order one with annual period. The implied ground-truth ROAS values, computed from the structural equation, are $\text{ROAS}_1 \approx 95$ and $\text{ROAS}_2 \approx 170$ on the original revenue scale. We compare four model variants: an oracle model (M1) that observes Z_t and establishes the achievable ceiling; a naive model (M2) that omits Z_t and uses uninformative priors on $(\alpha_k, \lambda_k, \beta_k)$; a reparameterized model (M3) that omits Z_t and parameterizes ROAS_k directly, with a LogNormal prior on ROAS_k built from aggregated experimental estimates following Wurm’s weighted-sum recipe (this is the setup of the public PyMC-Marketing notebook [PyMC Labs, 2024g]); and a calibrated model (M4) that omits Z_t , uses the same ROAS reparameterization, and additionally adds a per-experiment saturation-likelihood term (5) on each channel using lift estimates centered at the ground-truth ROAS values with $\sigma = 30$ on the original scale (the setup of PyMC Labs, 2024f). All models are fit in PyMC with NUTS via the `numpyro` backend.

5.2 ESTIMATION RESULTS

The naive model M2 inflates the channel-1 coefficient to $\hat{\beta}_1^{M2} = 1.485 [0.915, 2.099]$, roughly double the oracle value $\hat{\beta}_1^{M1} = 0.709 [0.262, 1.288]$, and absorbs the confounding contribution into the channel. On the unconfounded channel 2, the bias is small. Under M3, the aggregated-experiment prior pulls the ROAS_k posteriors onto the right scale: the channel-1 posterior is centered near 95 against ground truth 93.4 and the channel-2 posterior near 170 against ground truth 171.4 [PyMC Labs, 2024g]. Under M4, the per-experiment saturation likelihood tightens these posteriors further and concentrates them on ground truth, with credible intervals that contract progressively as more lift tests are incorporated [PyMC Labs, 2024f]. In this simulation M4 yields more accurate posteriors than M3, and both are directionally far closer to ground truth than M2, reversing the channel ranking that the naive fit would otherwise induce.

5.3 DECISION-LOSS EVALUATION

We compare four policies under a fixed total budget B : the oracle policy π^* under M1, the naive $\hat{\pi}^{M2}$, the reparameterized $\hat{\pi}^{M3}$, and the calibrated $\hat{\pi}^{M4}$. Each policy solves a constrained nonlinear program (SLSQP) maximizing posterior expected revenue subject to $\sum_k \pi_k = B$ and per-channel bounds; the objective is evaluated by pushing π through the posterior predictive of $S_{\lambda_k}(A_{\alpha_k}(\cdot))$, following the reference pipeline in PyMC-Marketing [PyMC Labs, 2024c]. The optimizer’s only structural inputs are the saturation parameters (α_k, λ_k) and the ROAS posterior, so the calibration recipe of Section 4 is what unlocks the optimization rather than a downstream concern: a biased posterior on these objects produces a biased argmax no matter how well-tuned the solver. Because M2 overstates channel 1’s effect, the naive policy over-allocates to the confounded channel; both M3 and M4 reverse this and reproduce the oracle’s ranking, with $\hat{\pi}^{M4}$ tracking the oracle most closely. Across a sweep of total budgets, M4 closes *approximately* $X\%$ of the revenue gap $V(\pi^*) - V(\hat{\pi}^{M2})$ between the naive and oracle policies and M3 closes a smaller but substantial fraction; the residual gap is attributable to weak identification of the saturation shape. The same expected-revenue objective extends to a risk-aware Bayesian objective that trades off mean response against tail risk in the posterior predictive (Value-at-Risk, or a mean-tightness criterion that penalizes posterior dispersion); tighter and unbiased posteriors on the saturation curves, the output of calibration, translate directly into better risk-adjusted allocations, not only better expected-value ones [PyMC Labs, 2024b]. Reporting in policy space, rather than coefficient space, aligns the metric with the decision-making premise: estimation error and decision error are not the same thing.

5.4 SENSITIVITY TO PRIOR MISSPECIFICATION

Lift tests are themselves biased by spillover, market heterogeneity, and staleness, so we probe robustness by sweeping the experimental prior mean across $\{0.5, 0.75, 1.0, 1.25, 1.5\} \times \text{ROAS}_k^{\text{true}}$ and the scale σ over a range, measuring decision loss (3) for each combination. The calibrated MMM is robust to $\pm 25\%$ prior misspecification at moderate σ and degrades substantially only when the prior is simultaneously biased and tight; the practical implication is to widen σ when experiments are few, stale, or known to be biased, rather than tightening it to chase a sharper ROAS posterior.

6 DISCUSSION

The recipe is a concrete instance of the data-fusion programme of Bareinboim and Pearl [2016], sharing its structural logic with the literatures on combining randomized

trials and observational studies in healthcare [Colnet et al., 2024, Kallus et al., 2018, Yang and Ding, 2020], off-policy evaluation [Dudík et al., 2014], and causal bandits [Lattimore et al., 2016]; what is specific to MMM is that the observational data is dense in spend levels but confounded while the experimental data is unconfounded but sparse and local, so the saturation-likelihood term is the natural way to let each experiment locate the curve at the spend pair it measured while the observational data shapes the rest. The argument is therefore not anchored on the simulation in Section 5: that simulation supplies a setting where ground truth is available to verify the bias correction, but the underlying problem (multicollinear co-activated channels with unobserved demand drivers) is a routine feature of production MMM deployments, where the same recipe applies whether or not an oracle is available to check against. The recipe is also not specific to the single-pooled-channel setting used here: the saturation-likelihood term plugs directly into geo-level hierarchical MMMs in the style of Sun et al. [2017], where lift tests in a subset of geos update population hyperparameters and propagate evidence to unobserved geos through the hierarchy; a worked PyMC-Marketing implementation reports roughly a 40% tightening of saturation posteriors at no cost to predictive fit [PyMC Labs, 2024d].

The limitations are as follows. First, the single confounder used in Section 5 is a didactic choice; both the ROAS reparameterization and the saturation-likelihood calibration extend without modification to multiple, non-stationary, and cross-geo dependent confounders. Second, lift experiments themselves can be biased: spillover, market heterogeneity, and pricing dynamics violate the unconfoundedness assumed in (5), and calibration with biased experiments propagates that bias [Coopridge and Nassiri, 2023]. Third, saturation and adstock parameters remain weakly identified even under calibration (in our M3 fit the 94% interval on λ_2 is $[0.073, 2.986]$ around a true value of 2.5); the remedy is experimental rather than structural, since additional lift tests at distinct spend levels collapse the ridge documented by Vincent [2026b]. Fourth, the decision analysis treats budget allocation as a single-period problem; sequential allocation under learning is the natural extension. Future work should connect this calibration scheme to causal bandits, hierarchical pooling across geographies and brands [Masoero et al., 2025], and debiasing priors for known-biased experiments.

7 CONCLUSION

Calibration via lift-test likelihood is a deployable instance of Bayesian data fusion: experimental observations of the saturation curve enter the same posterior as the observational time series, and the resulting model is parameterized in the quantity that the decision actually depends on. On a controlled simulation, calibration recovers the channel-level ROAS that a naive observational MMM badly biases, and

reverses the policy ranking that the bias would otherwise induce. Evaluating in policy space, rather than parameter space, makes the value of calibration visible in the units that matter (expected revenue), and is the right framing for any application whose ultimate output is a decision rather than a coefficient.

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